

NUMERACY NON-CALCULATOR



YEAR
9
2017

STUDENT: please print your name in the boxes below

FIRST NAME

LAST NAME

**DO NOT WRITE
IN THIS SPACE**

PART B

10 min

Time available for students to
complete Part B: 10 minutes

Use 2B
pencil **only**



YEAR 9 NUMERACY (NON-CALCULATOR)



- 1 There are 100 grams of raisins in a 500 gram packet of snack mix.

What percentage of the snack mix is made up of raisins?

- ☐ 0.05%
☐ 0.2%
☐ 2%
☐ 5%
☐ 20%
☐ 50%

- 2 The population of Mildura is approximately 48 000.

Which of these shows 48 000 in scientific notation?

4.8×10^3

☐

4.8×10^4

☐

4.8×10^5

☐

4.8×10^6

☐

- 3 Lydia is buying balloons for a party.

Each bag contains 8 balloons.

If she needs 100 balloons, how many bags will she need to buy?

- 4 Some motorbike engines run on a mixture of petrol and oil.

Directions for mixing the fuel for these engines are as follows:

**For every 2 litres of petrol
add 80 millilitres of oil.**

How many millilitres of oil should be added to 0.5 litres of petrol?

0.2 mL

☐

0.8 mL

☐

20 mL

☐

40 mL

☐

160 mL

☐

YEAR 9 NUMERACY (NON-CALCULATOR)



- 5** Bob's Bakery reports a profit of \$6405.70
 Karen's Bakery reports a profit of \$4070.95
 How much more money did Bob's Bakery make than Karen's Bakery?
 \$

- 6** $6 + (15 - 6 \times 2 + 1) = ?$
 What number does ? represent?

31

☐

25

☐

10

☐

8

☐

- 7** A store has 350 items for sale.
 Of the 350 items, $\frac{5}{7}$ of them cost more than \$21.
 Of the items for sale that cost more than \$21, $\frac{1}{5}$ cost more than \$30.
 How many items in the store cost more than \$30?

- 8** Kylie writes a pattern using indices.
 $1^1 \quad 2^2 \quad 3^3 \quad 4^4 \quad 5^5 \quad 6^6 \quad 7^7 \quad 8^8 \quad 9^9$
 What is the sum of the first four terms in Kylie's pattern?

$$1^1 + 2^2 + 3^3 + 4^4 =$$

STOP – END OF TEST

